

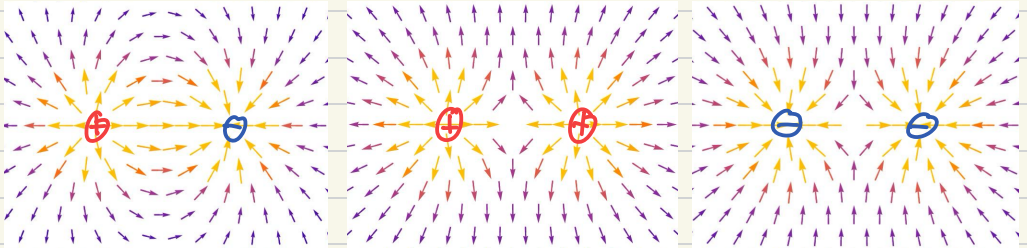
Vector Fields

April 20, 2026

def: A vector field on $\mathbb{R}^n$ is a mapping $\vec{F}: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$.

idea: A vector field assigns a vector in $\mathbb{R}^n$ to every point in X .

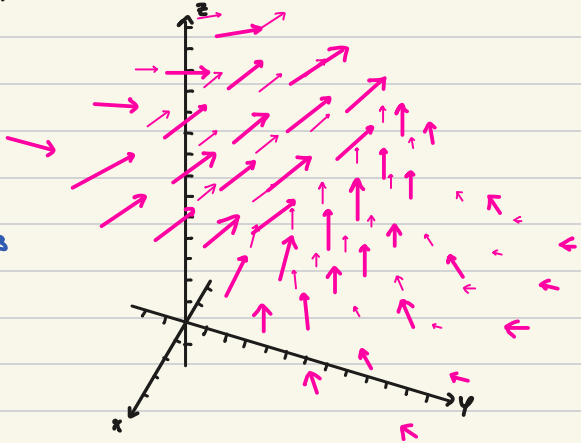
e.g. In $\mathbb{R}^2$, electric field produced by two charges placed on a surface



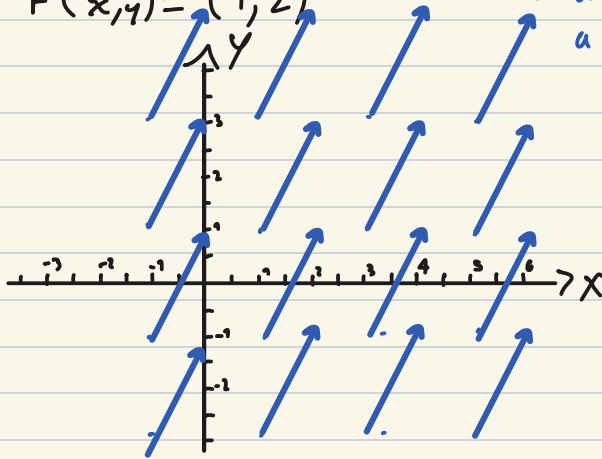
Can interpret this as the force vector a point-like positive charge would experience

e.g. Wind velocity at each point in $X \subseteq \mathbb{R}^3$

At each point in the air, the air is moving in some direction and at some speed which just looks like a vector in $\mathbb{R}^3$



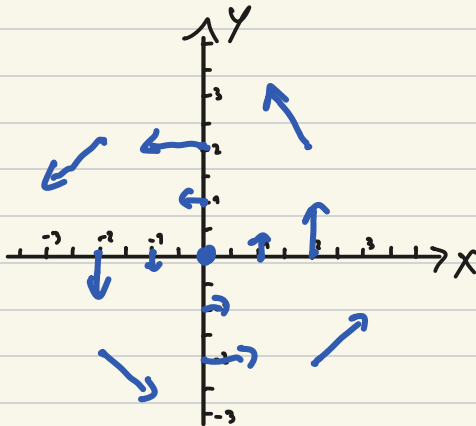
e.g. $\vec{F}(x,y) = (1, 2)$



constant, at every point we have a vector $(1, 2)$

e.g. $\vec{F}(x,y) = (-y, x)$

scale of vector 0.5



(x,y)	$\vec{F}(x,y)$
$(0,0)$	$(0,0)$
$(2,0)$	$(0,2)$
$(2,2)$	$(-2,2)$
$(0,2)$	$(-2,0)$

Rotational vector field

if we trace a path $\vec{x}(t) = (r \cos t, r \sin t)$, then $\vec{v}(t) = (-r \sin t, r \cos t)$ and $\vec{v}(t) = \vec{x}'(t) = \vec{F}(\vec{x}(t))$. fixed constant r and $0 \leq t \leq 2\pi$

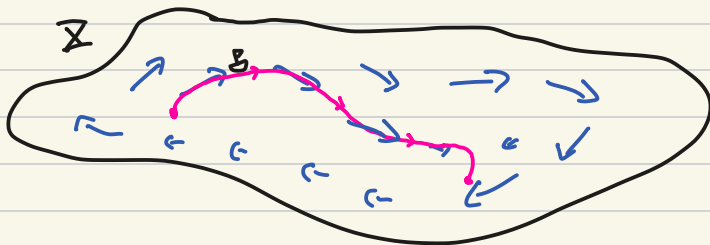
This vector field gives the velocity of some circular motion

(This will not be a crucial topic for us, but it is one of the main applications...)

Flow of a vector field

def: Suppose $\vec{F}: \overset{\text{open}}{\Sigma} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a vector field and $\vec{x}: I \rightarrow \Sigma \subseteq \mathbb{R}^n$ is a differentiable parametrized path. If $\vec{x}'(t) = \vec{F}(\vec{x}(t))$ for all $t \in I$, then we say $\vec{x}$ is a flow line of the vector field $\vec{F}$.

intuition: Imagine a vector field in $\mathbb{R}^2$ describes the water velocity in a lake and a boat is just floating and moving at the same speed of the water. The boat's path is a flow line

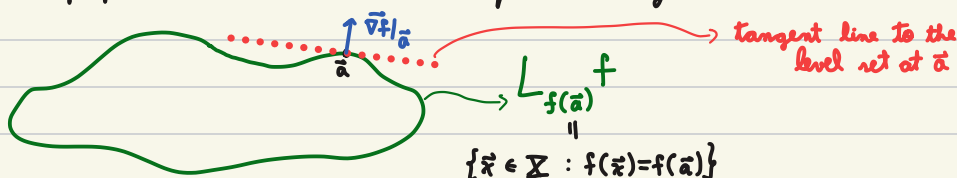


(... back to our main topic)

def: Suppose $\vec{F}: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a vector field. If there is a differentiable function $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\vec{\nabla} f(\vec{x}) = \vec{F}(\vec{x})$ for all $\vec{x}$ in X , then we say that:

- $\vec{F}$ is a gradient vector field (or conservative vector field),
- f is a potential function for the vector field $\vec{F}$.

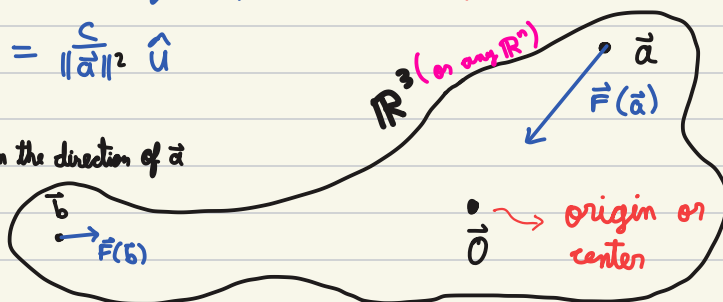
recall: Theorem (from Chapter 2): If $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is C^1 , then $\vec{\nabla} f|_{\vec{a}}$ is perpendicular to the level set of f containing $\vec{a}$



e.g. Inverse square law (gravitational and electrostatic forces)
vector field $\vec{F}: \mathbb{R}^3 \setminus \{0\} \rightarrow \mathbb{R}^3$ central force

$$\vec{F}(\vec{a}) = \frac{c}{\|\vec{a}\|^3} \vec{a} = \frac{c}{\|\vec{a}\|^2} \hat{u}$$

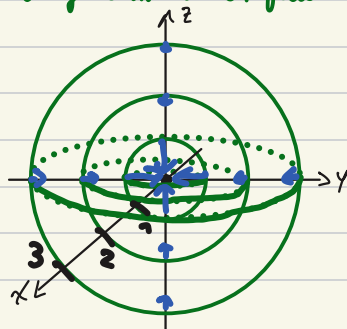
c is some constant,
 $\hat{u}$ is the unit vector in the direction of $\vec{a}$



Is a central force with inverse square law a gradient vector field?

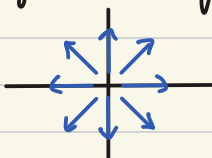
Sketch of

scale of v.f.
in sketch 1

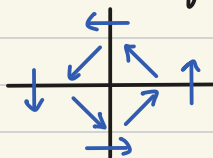


def: Suppose $\vec{F} = \vec{\nabla} f$ is a gradient vector field, the level sets of f are called equipotential sets.

We have seen two very common features of some vector fields in the examples:



"vector field is going in/out from a point"



"vector field is rotating around a point"

For vector fields in $\mathbb{R}^3$ (and $\mathbb{R}^2$) we can "measure how much rotation happens". We can "measure how much going in/out happens" in any vector field:

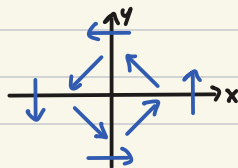
• def: Let $\vec{F}: \mathcal{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a differentiable vector field.

We say the divergence of $\vec{F}$ is the function $\nabla \cdot \vec{F}: \mathcal{X} \rightarrow \mathbb{R}$

$$\text{div}(\vec{F}) = \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \dots \quad (\text{i.e. the sum of the diagonal entries of } D\vec{F})$$

e.g. $\vec{G}(x, y) = (-y, x)$

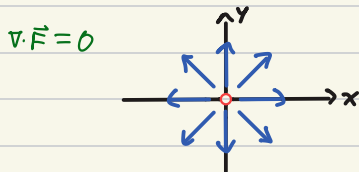
$$\nabla \cdot \vec{G} = \frac{\partial G_1}{\partial x} + \frac{\partial G_2}{\partial y} = \frac{\partial(-y)}{\partial x} + \frac{\partial(x)}{\partial y} = 0$$



e.g. Inverse square law central force

$$\vec{F}(\vec{x}) = \frac{c}{\|\vec{x}\|^3} \vec{x}$$

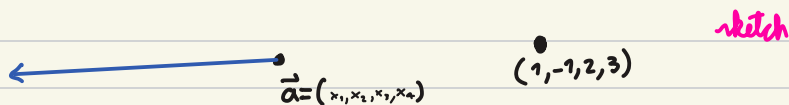
$$= \frac{c}{(\sqrt{x^2 + y^2 + z^2})^3} (x, y, z)$$



$$\begin{aligned} \text{div}(\vec{F}) &= \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{\partial}{\partial x} \left(\frac{cx}{(x^2 + y^2 + z^2)^{3/2}} \right) + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{c(x^2 + y^2 + z^2)^{-3/2} - cx \cdot \frac{3}{2}(x^2 + y^2 + z^2)^{-5/2} \cdot 2x}{(x^2 + y^2 + z^2)^3} \\ &= \frac{c}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3cx^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{c}{(x^2 + y^2 + z^2)^{3/2}} \left[3(x^2 + y^2 + z^2) - 3x^2 - 3y^2 - 3z^2 \right] + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ &= 0 \end{aligned}$$

e.g. $\vec{G}(x_1, x_2, x_3, x_4) = (x_1 - 1, x_2 + 1, x_3 - 2, x_4 - 3)$

$$\nabla \cdot \vec{G} = \frac{\partial G_1}{\partial x_1} + \frac{\partial G_2}{\partial x_2} + \frac{\partial G_3}{\partial x_3} + \frac{\partial G_4}{\partial x_4} = 1 + 1 + 1 + 1 = 4$$



Later, we will see how divergence measures how much a vector field is going out from a region (vs. how much is going in).

• def: Let $\vec{F}: \overset{\text{open}}{X} \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a differentiable vector field in $\mathbb{R}^3$, then the curl of $\vec{F}$ is the vector field $\nabla \times \vec{F}: X \rightarrow \mathbb{R}^3$

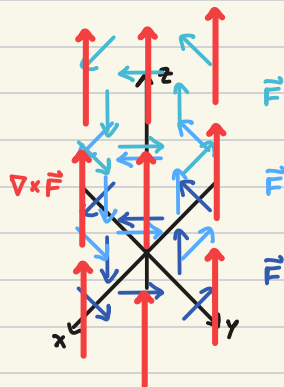
$$\text{curl}(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

This is not a regular matrix determinant, but rather a useful shorthand notation

$$= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \hat{k}$$

e.g. $\vec{F}(x, y, z) = (-y, x, 0)$

$$\begin{aligned} \text{curl} \vec{F} &= \nabla \times \vec{F} = \\ &= 0\hat{i} + 0\hat{j} + \left(\frac{\partial F_1}{\partial x} - \frac{\partial F_2}{\partial y} \right) \hat{k} \\ &= 0\hat{i} + 0\hat{j} + (1 - (-1))\hat{k} = 2\hat{k} \end{aligned}$$



* We can "measure rotation" for vector fields in $\mathbb{R}^3$ by looking at scalar curl in $\mathbb{R}^2$ $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$

e.g. $\vec{F}(x,y,z) = (x^2y + 4z, x \cos(y), e^{xz} + y)$

$$\begin{aligned}\nabla \times \vec{F} &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) \hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right) \hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) \hat{k} \\ &= (1 - 0) \hat{i} + (4 - ze^{xz}) \hat{j} + (\cos(y) - x^2) \hat{k} \\ &= 1 \hat{i} + (4 - ze^{xz}) \hat{j} + (\cos(y) - x^2) \hat{k}\end{aligned}$$

the letter ∇ is called nabla

notation/def: The book defines the **del operator** (denoted ∇) in Cartesian coordinates in $\mathbb{R}^3$ as

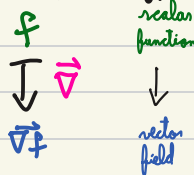
$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

This ∇ operator acts on functions on $\mathbb{R}^3$ (scalar or $\mathbb{R}^3$ vector valued):

• If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is a differentiable scalar function, then

∇ acts on f as the gradient:

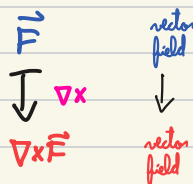
$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = \vec{\nabla} f$$



• If $\vec{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a differentiable vector field, then

$\nabla \times$ acts on $\vec{F}$ as the curl:

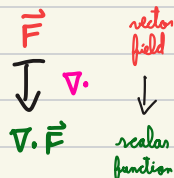
$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$



• If $\vec{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a differentiable vector field, then

$\nabla \cdot$ acts on $\vec{F}$ as the divergence:

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$



Theorem ("curl of gradient"): Let $f: X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ be a C^2 function (all 1st and 2nd order partial derivatives exist and are continuous), then $\text{curl}(\nabla f) = \vec{0}$, i.e. $\nabla \times \nabla f = \vec{0}$.

Theorem ("divergence of curl"): Let $\vec{F}: X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a C^2 vector field (all 1st and 2nd order partial derivatives exist and are continuous), then $\text{div}(\text{curl } \vec{F}) = 0$, i.e. $\nabla \cdot \nabla \times \vec{F} = 0$.